of the sign. Suppose, we are given two logical variables such that A=B, if A=1, then B=1 and if A=0, then B=0.

2. Plus Sign (+): The plus sign in Boolean algebra refers to the logical OR operation. Thus, when the statement A+B=1 appears in Boolean algebra, it means A ORed with B equals1.Consequently, either A=1 or B=1 or both equals1.

3. Multiply Sign (.): The multiply sign in Boolean algebra refers to AND operation .Thus, when the statement A.B=1 appears in Boolean algebra, it means A ANDed with B equals 1. Consequently, A=1 and B=1. The function A.B is often written as AB, omitting the dot for convenience.

4. Bar sign (-): The bar sign in Boolean algebra refers to NOT operation. The NOT has the effect of inverting (complementing) the logical value. Thus, if A=1, then $\overline{A}=0$.

8.8.1 Boolean Theorems

We now discuss the basic Boolean theorems that are useful in manipulating and simplifying Boolean expression. For convenience, we divide the theorems into two groups:

1. Single variable theorem
2. Multi variable theorem

Single Variable Theorems: These theorems refer to the condition when only one input of the logic gate is variable. Table 8.6 gives single variable Boolean theorems.

Table 8.6: Single Variable Boolean Theorems

Name of the Theorem	Mathematical Function
Theorem 1	$A + 0 = A$
Theorem 2	$A \cdot 1 = A$
Theorem 3	$A + \overline{A} = 1$
Theorem 4	$A \cdot \overline{A} = 0$
Theorem 5	$A + A = A$
Theorem 6	$A \cdot A = A$
Theorem 7	$A + 1 = 1$
Theorem 8	$A \cdot 0 = 0$
Theorem 9	$\overline{\overline{A}} = A$

Theorem 1 (A+0=A): This theorem can be verified by ORing a variable A with a 0 and is illustrated in Figure 8.3. Here, one input to OR gate is always 0 and the other input A can be a value 1 or 0. When A is at 1, the output is 1 which is equal to A. When A is at 0, the output is 0 which is also equal to A(=0). Therefore, a variable ORed with 0 is equal to the value of the variable. This is easy to remember, since 0 added to anything does not affect the value of the variable, either in regular addition or OR addition.

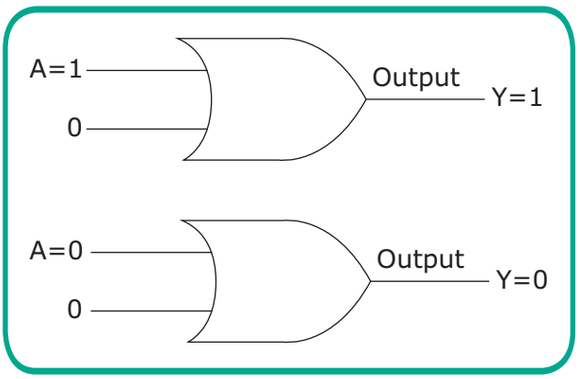


Figure 8.3 Logic gate representation of Theorem 1

Theorem 2(A.1=A): This theorem can be verified by ANDing a variable A with a 1 and is illustrated in Figure 8.4. Here, one input to AND gate is always 1

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